Continuity in Real Analysis

Definition of Continuity

A function $f : D \to \mathbb{R}$ is continuous at $c \in D$ if:

- For every neighborhood V of $f(c)$
- There exists a neighborhood W of c
- Such that $f(x) \in V$ for all $x \in W \cap D$

Equivalent ϵ-δ definition: A function $f : D \to \mathbb{R}$ is continuous at $c \in D$ if: $lim_{x \to c} f(x) = f(c)$ i.e.,

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } |f(x) - f(c)| < \epsilon \text{ for all } x \in N(c, \delta) \cap D$$

Key Theorems on Continuity

Theorem (Sequential Criterion)

f is continuous at $c \in D \cap D'$ iff for every sequence $\{x_n\}$ in D converging to c, the sequence $\{f(x_n)\}$ converges to $f(c)$.

Theorem (Algebra of Continuous Functions)

If f and g are continuous at c, then:

- *$f + g$ is continuous at c*
- *kf is continuous at c ($k \in \mathbb{R}$)*
- *fg is continuous at c*
- *f/g is continuous at c (if $g(c) \neq 0$)*

Composite Functions and Continuity

Theorem (Continuity of Composites)

If:

- f *is continuous on* D
- g *is continuous on* $f(D)$

Then $g \circ f$ *is continuous on* D*.*

Examples:

- $\sqrt{f(x)}$ is continuous if $f(x) \geq 0$ and f continuous
- $\log f(x)$ is continuous if $f(x) > 0$ and f continuous
- $e^{f(x)}$ is continuous if f is continuous

Classification of Discontinuities

- **Removable discontinuity**: $\lim_{x \to c} f(x)$ exists but $\neq f(c)$
- **Jump discontinuity**: $f(c+0)$ and $f(c-0)$ exist but are unequal
- **Oscillatory discontinuity**: $\lim_{x \to c} f(x)$ doesn't exist but f is bounded nearby
- **Infinite discontinuity**: f is unbounded in every neighborhood of c

Examples of Discontinuous Functions

Example (Dirichlet Function)

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \in \mathbb{R} - \mathbb{Q} \end{cases}$$

Discontinuous at every point.

Example (Removable Discontinuity)

$$f(x) = \frac{x^2 - x}{x^2 + x}, x > 2; \quad f(2) = 10$$

Discontinuous at $x = 2$ (removable).

Theorem 4.0

Theorem

Let $D \subset \mathbb{R}$ *and a function* $f : D \to \mathbb{R}$ *be continuous at* $c \in D$. *Then* $|f|$ *is continuous at* c.

Proof of Theorem 4.0

Proof: Define $|f| : D \to \mathbb{R}$ by $|f|(x) = |f(x)|$, $x \in D$.

$$\big|\, |f(x)| - |f(c)| \,\big| \leq |\, f(x) - f(c) \,|$$

Let $\varepsilon > 0$. Since f is continuous at c, $\exists\, \delta > 0$ such that

$$|f(x) - f(c)| < \varepsilon \quad \forall x \in N(c, \delta) \cap D$$

Therefore,

$$\big|\, |f(x)| - |f(c)| \,\big| < \varepsilon \quad \forall x \in N(c, \delta) \cap D$$

Thus, $|f|$ is continuous at c.

Theorem 4.1

Theorem

Let $D \subset \mathbb{R}$ and $f : D \to \mathbb{R}$ be continuous on D. Then $|f|$ is continuous on D.

Note on Theorem 4.1

Note:

If $|f|$ is continuous on D, then f may not be continuous on D.

Example:

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ -1, & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

Then $|f|(x) = 1$ for all $x \in \mathbb{R}$.
Here $|f|$ is continuous on $\mathbb{R}$, but f is not continuous on $\mathbb{R}$.

Fundamental Theorems of Continuous Functions

Theorem (Bolzano's Theorem)

If f is continuous on $[a, b]$ and $f(a)f(b) < 0$, then $\exists c \in (a, b)$ such that $f(c) = 0$.

$$f(x) = x^2 - 2, x \in [1, 2]$$

$$f(1) = -1 < 0, f(2) = 2 > 0$$

$\Rightarrow c = \sqrt{2} \in (1, 2)$ such that $f(\sqrt{2}) = 0$

Intermediate Value Theorem

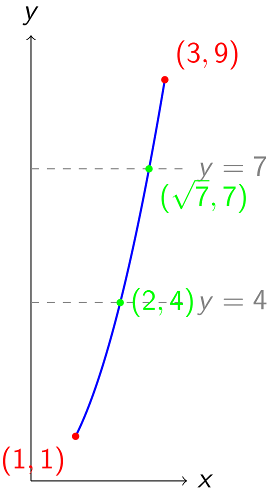
Theorem

If f is continuous on $[a, b]$ and $f(a) \neq f(b)$, then for every y between $f(a)$ and $f(b)$,

$$\exists c \in (a, b) \quad \text{such that } f(c) = y.$$

- Let $f(x) = x^2$ on $[1, 3]$.
- $f(1) = 1$, $f(3) = 9$.
- By IVT, f attains every value in $(1, 9)$.
- E.g. $f(2) = 4$, $f(\sqrt{7}) = 7$.

Example with Graph



Proof of IMV Theorem

Proof: Without loss of generality, assume $f(a) < f(b)$. Let μ be a real number such that $f(a) < \mu < f(b)$.
Define $\phi : [a, b] \to \mathbb{R}$ by $\phi(x) = f(x) - \mu$.

$$\phi(a) = f(a) - \mu < 0, \quad \phi(b) = f(b) - \mu > 0$$

Since ϕ is continuous, by Bolzano's theorem there exists $c \in (a, b)$ such that $\phi(c) = 0$.
Thus $f(c) = \mu$. Hence, f attains every value between $f(a)$ and $f(b)$ in (a, b).

Note

Let $I = [a, b]$ be a closed and bounded interval and $f : [a, b] \to \mathbb{R}$ be such that

$$f(a) \neq f(b)$$

and f attains every value between $f(a)$ and $f(b)$ at least once in (a, b).

Still, f may not be continuous on $[a, b]$.

Example

Let $f : [0, 2] \to \mathbb{R}$ be defined by

$$f(0) = 0, \quad f(2) = 2$$

and

$$f(x) = \begin{cases} x, & 0 < x \leq 1, \\ 3 - x, & 1 < x < 2. \end{cases}$$

Then f assumes every value between 0 and 2 on $[0, 2]$. The function f is not continuous on $[0, 2]$, since f is not continuous at $x = 1$ and $x = 2$.

Extreme Value and Image Theorems

Theorem (Extreme Value Theorem)

Let $I = [a, b]$ be a closed and bounded interval and a function $f : I \to \mathbb{R}$ be continuous on I. Then:

1. *f is bounded on $[a, b]$*
2. *$\exists c, d \in [a, b]$ such that $f(c) = \sup f$ and $f(d) = \inf f$*

Continuous Image Theorem

Theorem

Let I be an interval and $f : I \to \mathbb{R}$ be continuous on I. Then $f(I)$ is an interval.

(Definition:) A subset S of $\mathbb{R}$ is an interval if for any $c, d \in S$ with $c < d$, the closed interval $[c, d] \subseteq S$.

Proof of Theorem

Proof: Let $p, q \in f(I)$ with $p < q$. Then there exist $c, d \in I$ such that $f(c) = p, f(d) = q$.
Let $r \in (p, q)$. Then $p < r < q$. By the Intermediate Value Theorem, $\exists x_0 \in (c, d)$ such that $f(x_0) = r$.
Thus $r \in f(I)$, hence $(p, q) \subset f(I)$. Also $p, q \in f(I) \implies [p, q] \subset f(I)$.
Therefore, $f(I)$ is an interval.

Theorem (Fixed Point Theorem)

If $f : [a, b] \to [a, b]$ is continuous, then $\exists c \in [a, b]$ such that $f(c) = c$.

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Proof Idea

Define

$$g(x) = f(x) - x, \quad x \in [a, b].$$

- At $x = a$: $g(a) = f(a) - a \geq 0$.
- At $x = b$: $g(b) = f(b) - b \leq 0$.

Since g is continuous on $[a, b]$, by the Intermediate Value Theorem, $\exists c \in [a, b]$ such that

$$g(c) = 0 \quad \Rightarrow \quad f(c) = c.$$

Uniform Continuity

- Continuity ensures that small changes in x lead to small changes in $f(x)$.
- However, the *size of the neighborhood* may depend on the point x.
- Example: $f(x) = 1/x$ on $(0, 1)$ is continuous at every point, but as $x \to 0$, we need smaller and smaller neighborhoods.
- Question: Can we control the variation of $f(x)$ *uniformly* for all x in the domain?

Uniform Continuity

Definition

$f : I \to \mathbb{R}$ is uniformly continuous on I if:

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } |x_1 - x_2| < \delta \Rightarrow |f(x_1) - f(x_2)| < \epsilon$$

for all $x_1, x_2 \in I$.

- Difference from continuity: For ordinary continuity, δ may depend on the point c. For uniform continuity, the same δ works for *all points* in D.

Examples

Example 1: $f(x) = x^2$ on $[0, 1]$.

$$|x^2 - y^2| = |x - y||x + y| \leq 2|x - y|, \quad \forall x, y \in [0, 1].$$

Thus, f is uniformly continuous on $[0, 1]$.

Example 2: $f(x) = 1/x$ on $(0, 1)$.

$$|f(x) - f(y)| = \left|\frac{y - x}{xy}\right|$$

Near $x = 0$, the denominator becomes very small, so no single δ works for all points. Hence, f is not uniformly continuous on $(0, 1)$.

Heine–Cantor Theorem

Theorem: Every continuous function on a closed and bounded interval $[a, b]$ is uniformly continuous.

Concept

- A function may be continuous everywhere on its domain but still fail to be uniformly continuous.
- This typically happens on unbounded intervals.

Example 1: $f(x) = x^2$ on $(0, \infty)$

- $f(x) = x^2$ is continuous on $(0, \infty)$.
- For $x, y \in (0, \infty)$:

$$|f(x) - f(y)| = |x^2 - y^2| = |x - y||x + y|$$

- Even if $|x - y|$ is small, the factor $|x + y|$ can be arbitrarily large as $x, y \to \infty$.
- Hence, no single δ works for all x, y. $\Rightarrow$ Not uniformly continuous.

Uniform Continuity on $[1, \infty)$

Problem: Show that the function

$$f(x) = \frac{1}{x}, \quad x \in [1, \infty)$$

is uniformly continuous on $[1, \infty)$.

Proof (Step 1)

Let $c \geq 1$. Then for all $x \geq 1$,

$$|f(x) - f(c)| = \left|\frac{1}{x} - \frac{1}{c}\right| = \left|\frac{x - c}{cx}\right|.$$

Since $|cx| \geq 1$ (because $c, x \geq 1$), we have

$$|f(x) - f(c)| \leq |x - c|.$$

Proof (Step 2)

Now, let $\varepsilon > 0$. Choose $\delta = \varepsilon$.
Then, for all $x \geq 1$, if $|x - c| < \delta$, we get

$$|f(x) - f(c)| \leq |x - c| < \varepsilon.$$

Thus, the choice of δ is independent of c.

Thank You